

Duca Raul

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Languages: English (C1), Romanian (Native)

ABOUT

Final-year Mathematics and Computer Science student passionate about web development, game development, and cybersecurity. Fascinated by the cosmos and space exploration.

EDUCATION

Babeş-Bolyai University, Faculty of Mathematics and Computer Science

2023 – 2026

Bachelor in Mathematics (Mathematics-Informatics)

Last Semester GPA: 9.36/10

- Relevant coursework: Data Structures, Object-Oriented Programming, Web Programming, Algorithms and Programming, Databases, Software Engineering, Advanced Programming Methods, Computer Networks, Operating Systems, Human-Computer Interaction, Computer Architecture.

TECHNICAL SKILLS

Front-End : Angular, Next.js, HTML, CSS, JavaScript, TypeScript, Kivy

Back-End : Spring Boot, Node.js, Java, C++, Python, SQL

Mathematical & Scientific Computing : Matlab, Maple

Developer Tools : Git, Docker, PostgreSQL, Swagger, Flyway, WebSockets, RabbitMQ, JWT

Methodologies : Agile (Scrum), OOP, MVC, Layered Architecture, Design Patterns

EXPERIENCE & PERSONAL PROJECTS

Pet Adoption App | Spring Boot (Java), Angular (TypeScript), PostgreSQL, Docker, Git, Swagger Aug 2025 – Oct 2025

- Developed in a team during the **Training programme at Cloudflight in partnership with UBB**, following **Agile** and **Scrum** methodologies.
- Implemented **RESTful backend** using **Spring Boot** with database migration tools (**Flyway**) and integrated **PostgreSQL** via **Docker**.
- Built **responsive frontend** components in **Angular** using **NgRx** for state management, implemented user profile, pet listing, pet filtering and adoptions.

HomeMate - Household Management App | Spring Boot, Angular, JWT, RabbitMQ

Oct 2025 – Jan 2026

- Collaborated in a team to build a **full-stack web application** for household management, contributing across the stack with a primary focus on **frontend development**.
- Developed the responsive and intuitive UI using **Angular** and **TypeScript**, managing features such as rooms, tasks, dashboards, and user profiles.
- Backend built with **Spring Boot** following a **Layered Architecture**, utilizing **JWT** for security and **RabbitMQ** for publish-subscribe messaging.

Campionii României - Educational Game | Python, Kivy

Oct 2025 – Jan 2026

- Collaborated within a team to develop the **frontend** of an educational **Android** tablet game aimed at 4th-grade students to facilitate the learning of geography and history, using **Python** and **Kivy**.
- Implemented interactive mini-games, quizzes, timers, and a map interface divided into regions.
- Focused on **HCI (Human-Computer Interaction) principles**, creating an intuitive, accessible, and highly visual UI with integrated reward systems.

Competitive Wordle 1vs1 | Next.js, Node.js, WebSockets

2026

- Developing a **competitive multiplayer version** of the popular game Wordle, playable over a local area network.
- Built a fast, modern frontend interface using **Next.js** and managed the game logic and real-time synchronization with a **Node.js** backend.